Supplementary material for "CLAMP-RT: Falsification-Robust Fairness Control for Federated Graph Learning via a Two-Server Trimmed Statistic with Attribute-Level Differential Privacy"

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Section S1 gives complete proofs of the results of Section 5 of the main text; theorem numbers follow the main text. Section S2 contains the full tables of the attack studies and the privacy sweep. All tables are generated by `make_tables.py` from the result files listed in its header.

S1 Proofs

Notation For $x \in \mathbb{R}^K$, $x_{(1)} \le \cdots \le x_{(K)}$ are its sorted entries and $\mathrm{TM}_b(x) = \frac{1}{K-2b}\sum_{i=b+1}^{K-b} x_{(i)}$ for $0 \le b$ with $K - 2b \ge 1$. For a validation set $\mathcal{V}_k$ split into groups $\mathcal{V}_{k,0}, \mathcal{V}_{k,1}$ by the sensitive attribute, $m_{k,a} = \sum_{v\in\mathcal{V}_{k,a}} p(v)/\max(|\mathcal{V}_{k,a}|, n_{0,k})$ with $n_{0,k} = \max(1, \lceil c|\mathcal{V}_k|\rceil)$, and $d_k = \mathrm{clip}(m_{k,1} - m_{k,0}, -1, 1)$.

Lemma 1 (order statistics)

Claim (a). The map $x \mapsto x_{(i)}$ is non-decreasing in every entry and satisfies $(x + t\mathbf{1})_{(i)} = x_{(i)} + t$.

Proof. We have $x_{(i)} = \max_{S\subseteq[K],|S|=K-i+1} \min_{l\in S} x_l$: for the set S of indices of the $K - i + 1$ largest entries the minimum is $x_{(i)}$, and any set of $K - i + 1$ indices contains an index among the i smallest entries, so its minimum is at most $x_{(i)}$. A maximum of minima of coordinates is non-decreasing in every coordinate and commutes with adding a constant to all coordinates. □

Claim (b). If x, y differ only in entry j, then all differences $y_{(i)} - x_{(i)}$ have the sign of $y_j - x_j$ (or vanish) and $\sum_i |y_{(i)} - x_{(i)}| = |y_j - x_j|$.

Proof. Suppose $y_j \ge x_j$ (the other case is symmetric). Then $y \ge x$ entrywise, so $y_{(i)} \ge x_{(i)}$ for every i by (a). Sorting is a permutation, so $\sum_i y_{(i)} = \sum_i y_i$ and $\sum_i x_{(i)} = \sum_i x_i$; hence $\sum_i |y_{(i)} - x_{(i)}| = \sum_i (y_{(i)} - x_{(i)}) = y_j - x_j$. □

Claim (c). If x, y differ only in entry j, then $|\mathrm{TM}_b(x) - \mathrm{TM}_b(y)| \le |x_j - y_j|/(K - 2b)$.

Proof. $|\mathrm{TM}_b(x) - \mathrm{TM}_b(y)| \le \frac{1}{K-2b}\sum_{i=b+1}^{K-b} |x_{(i)} - y_{(i)}| \le \frac{1}{K-2b}\sum_{i=1}^{K} |x_{(i)} - y_{(i)}|$, and (b). □

Claim (d). For any $x, y \in \mathbb{R}^K$, $\max_i |x_{(i)} - y_{(i)}| \le \|x - y\|_\infty$, and $|\mathrm{TM}_b(x) - \mathrm{TM}_b(y)| \le \|x - y\|_\infty$.

Proof. Let $\epsilon = \|x - y\|_\infty$. Then $y \le x + \epsilon\mathbf{1}$ and $x \le y + \epsilon\mathbf{1}$ entrywise, so by (a) $y_{(i)} \le x_{(i)} + \epsilon$ and $x_{(i)} \le y_{(i)} + \epsilon$. Averaging over kept indices gives the second statement. □

Lemma 2 (sensitivity of a client's gap)

Claim. Fix the predictions $p(v) \in [0, 1]$ of client k's validation nodes. If the sensitive attribute of one validation node changes, d_k changes by at most $2/n_{0,k}$.

Proof. Write $n_0 = n_{0,k} \ge 1$. Let the node have prediction p, and consider the group it joins, with $N \ge 0$ nodes and mass $S = \sum p(v) \in [0, N]$ before the change. Its rate changes from $S/\max(N, n_0)$ to $(S + p)/\max(N + 1, n_0)$. Since N and n_0 are integers, either $N + 1 \le n_0$ or $N \ge n_0$.

- If $N + 1 \le n_0$, both denominators equal n_0, and the change is $p/n_0 \le 1/n_0$.

- If $N \geq n_0 \geq 1$, the change is $\frac{S+p}{N+1} - \frac{S}{N} = \frac{pN-S}{N(N+1)}$. Since $0 \leq pN \leq N$ and $0 \leq S \leq N$, $|pN - S| \leq N$, and the change is at most $1/(N+1) \leq 1/n_0$ in absolute value.

The group the node leaves goes from N' nodes with mass S' to $N'-1$ nodes with mass $S'-p$; this is the computation above with the roles of the two states exchanged (apply it to the group of $N'-1$ nodes that the node joins), so the change is again at most $1/n_0$. The other group's rate and the predictions are unchanged. Hence $|\Delta(m_{k,1} - m_{k,0})| \leq 2/n_0$, and clipping to $[-1,1]$ is 1-Lipschitz. □

Theorem 3 (attribute-level differential privacy)

Setting. Training lasts T rounds and statistics are released in the $R = \lceil T/P \rceil$ rounds t with $t \bmod P = 0$. Neighbouring datasets D, D' differ only in the attribute of one validation node v of one honest client k. We fix the training data of all clients, the initialisation and all randomness of local training; Byzantine clients' submissions are arbitrary (possibly randomised) functions of public information, which includes all released values, and of their own data. At least one server, say A, is honest. The released values are $\tilde{g}^{(j)} = \mathrm{TM}_b(\tilde{d}^{(j)}) + \xi_A^{(j)} + \xi_B^{(j)}$, $j = 1, \ldots, R$, with independent $\xi_A^{(j)} \sim \mathcal{N}(0, \sigma_d^2)$, and the update sums U^t.

Step 1: the update channel does not depend on s_v given the releases. The model θ^t is determined by θ^{t-1}, $\lambda^{t-1}, \omega^{t-1}$ and the accepted updates. Each honest client's update depends on θ^{t-1}, $(\lambda^{t-1}, \omega^{t-1})$, its training data and its local randomness; the local objective uses training nodes only, and the backbone does not read s. Byzantine updates depend on public information. (λ, ω) are functions of the released gaps. By induction over rounds, every model, update and acceptance decision is the same function of the release history under D and D'. In particular, given the history of the first $j-1$ releases, the inputs to the j-th release differ between D and D' only in d_k, by at most $2/n_{0,k} \leq 2/n_{0,\min}$ (Lemma 2), where we used that the predictions are identical. Honest client k is accepted in both cases ($d_k \in [-1,1]$ and the norm check does not involve s_v), so the vectors $\tilde{d}^{(j)}$ differ in one entry.

Step 2: one release. By Lemma 1(c), $|\mathrm{TM}_b(\tilde{d}^{(j)}(D)) - \mathrm{TM}_b(\tilde{d}^{(j)}(D'))| \leq \Delta_{\mathrm{TM}} = 2/(n_{0,\min}(K - 2b))$. Consider the adversary's view in the worst case that server B is curious: it knows $\xi_B^{(j)}$ and therefore observes $\mathrm{TM}_b + \xi_A^{(j)}$. By the security of the two-party protocol against a semi-honest server, B's view of the computation is simulatable from its own input and the opened values, so it learns nothing about $\tilde{d}$ beyond $\mathrm{TM}_b + \xi_A$ (and U^t, which by Step 1 adds nothing that depends on s_v given the history). The map $D \mapsto \mathrm{TM}_b(\tilde{d}^{(j)}(D)) + \xi_A^{(j)}$ is a Gaussian mechanism with sensitivity Δ_{TM} and standard deviation $\sigma_d = z\Delta_{\mathrm{TM}}$, which is $(\alpha, \alpha\Delta_{\mathrm{TM}}^2/(2\sigma_d^2)) = (\alpha, \alpha/(2z^2))$-RDP for every $\alpha > 1$ [Mironov 2017]. If both servers are honest, the observer sees noise of variance $2\sigma_d^2$, and the guarantee is stronger; we use the weaker one.

Step 3: composition. The j-th release is a Gaussian mechanism conditionally on the history of the earlier releases (Step 1), so adaptive composition of RDP over R releases gives $(\alpha, R\alpha/(2z^2))$-RDP, and the conversion of Canonne, Kamath and Steinke gives (ε, δ)-DP with $\varepsilon = \min_{\alpha>1}[R\alpha/(2z^2) + \log\frac{\alpha-1}{\alpha} - \frac{\log\delta + \log\alpha}{\alpha-1}]$. The code computes z by bisection on this expression over a fixed grid of orders. The bias-corrected magnitude $\widehat{D}$, λ, ω, the stored model and the returned model are functions of the releases and of quantities that by Step 1 do not depend on s_v given the releases, hence post-processing. □

What is not covered. The proof treats the noise as continuous and the trimmed mean as exact. A deployment over $\mathbb{Z}_{2^{64}}$ computes the trimmed mean of fixed-point values and must sample discrete noise; the sensitivity then increases by the rounding error and the RDP of a sum of two discrete Gaussians must be bounded separately (Canonne et al.; Kairouz et al.). The proof also assumes a semi-honest server.

Theorem 4 (update influence)

Let A be the accepted set, $B \subseteq A$ the accepted Byzantine clients with $|B| = f'$, $H = A \setminus B$, and suppose every honest client is accepted, so $|H| \geq K - f$. With $\bar{\Delta}_H, \bar{\Delta}_B$ the means over H and B, $U/|A| = \frac{|H|}{|A|}\bar{\Delta}_H + \frac{f'}{|A|}\bar{\Delta}_B$, so $U/|A| - \bar{\Delta}_H = \frac{f'}{|A|}(\bar{\Delta}_B - \bar{\Delta}_H)$. Every accepted update has norm at most C, so $\|\bar{\Delta}_B - \bar{\Delta}_H\|_2 \leq 2C$, and $|A| \geq |H| \geq K - f$, $f' \leq f$. □

Theorem 5 (falsified gaps)

Let $f \leq b$ Byzantine clients submit arbitrary values; values outside $[-1, 1]$ are replaced by 0, so every entry of $\tilde{d}$ lies in $[-1, 1]$, and the $K-f$ honest entries are the honest gaps $h_{(1)} \leq \cdots \leq h_{(K-f)}$. Fix i with $b + 1 \leq i \leq K - b$.

- Lower bound. The entries $\tilde{d}_{(1)}, \ldots, \tilde{d}_{(i)}$ are i entries $\leq \tilde{d}_{(i)}$; at most f of them are Byzantine, so at least $i - f$ honest gaps are $\leq \tilde{d}_{(i)}$. Since $i - f \geq b + 1 - f \geq 1$, the $(i - f)$-th smallest honest gap satisfies $h_{(i-f)} \leq \tilde{d}_{(i)}$.
- Upper bound. The entries $\tilde{d}_{(i)}, \ldots, \tilde{d}_{(K)}$ are $K - i + 1$ entries $\geq \tilde{d}_{(i)}$; at least $K - i + 1 - f$ of them are honest. Since $i \leq K - b \leq K - f$, this number is at least 1, and the $(K - i + 1 - f)$-th largest honest gap, which is $h_{(i)}$, satisfies $h_{(i)} \geq \tilde{d}_{(i)}$.

Averaging $h_{(i-f)} \leq \tilde{d}_{(i)} \leq h_{(i)}$ over $i = b + 1, \ldots, K - b$ gives the two inner inequalities of Eq. (6) of the main text; the outer ones follow from $h_{(i-f)} \geq h_{(b+1-f)}$ and $h_{(i)} \leq h_{(K-b)}$ for kept i. □

Consequences. (i) is immediate. (ii) If all honest gaps except at most $b - f$ exceed τ', then $h_{(b-f+1)} > \tau'$ and so $\mathrm{TM}_b(\tilde{d}) > \tau'$; on the event $|\xi_A + \xi_B| \leq r$ the released $\tilde{g}$ exceeds $\tau' - r$. (iii) $\mathrm{TM}_b(\tilde{d})$ and the analogous honest-only quantity both lie in $[h_{(b-f+1)}, h_{(K-b)}]$.

Breakdown. If $f = b + 1$ and all liars report -1, the smallest $b + 1$ entries are -1 (when no honest gap equals -1), one of them is kept, and TM_b is the average of -1 and $K - 2b - 1$ honest values, which can lie outside the honest range. Each kept liar value can shift TM_b by up to $2/(K - 2b)$ relative to an honest value.

Theorem 6 (returned model)

For $\zeta \sim \mathcal{N}(0, s^2)$, $\Pr[|\zeta| > r] \leq 2\exp(-r^2/(2s^2))$. With $s^2 = 2\sigma_d^2$ and $r = r_R = 2\sigma_d\sqrt{\ln(2R/\beta)}$ the right-hand side is $2\exp(-\ln(2R/\beta)) = \beta/R$. A union bound over the R independent releases gives $\Pr[\forall j : |\tilde{g}^{(j)} - \mathrm{TM}_b(\tilde{d}^{(j)})| \leq r_R] \geq 1 - \beta$. A stored model satisfies $\widehat{D} \leq \tau$, which by the definition $\widehat{D} = \sqrt{\max(0, \tilde{g}^2 - 2\sigma_d^2)}$ is equivalent to $\tilde{g}^2 \leq \tau^2 + 2\sigma_d^2$, i.e. $|\tilde{g}| \leq \tau_{\mathrm{eff}}$. On the event, $|\mathrm{TM}_b(\tilde{d}(\hat{\theta}))| \leq |\tilde{g}| + r_R \leq \tau_{\mathrm{eff}} + r_R$. If $f \leq b$, Theorem 5 gives $h_{(b-f+1)} \leq \mathrm{TM}_b \leq h_{(K-b)}$, hence $h_{(b-f+1)}(\hat{\theta}) \leq \tau_{\mathrm{eff}} + r_R$ and $h_{(K-b)}(\hat{\theta}) \geq -(\tau_{\mathrm{eff}} + r_R)$. Note that the event concerns the noise only, so the bound holds whatever the liars do. □

Proposition 7 (validation to population)

Under the stated assumption, conditionally on the group sizes $n_{k,a} \geq n_{0,k}$, $m_{k,a}$ is the mean of $n_{k,a}$ independent values in $[0, 1]$ with mean $\mu_{k,a}$, so Hoeffding's inequality gives $\Pr[|m_{k,a} - \mu_{k,a}| > e] \leq 2\exp(-2n_{k,a}e^2)$. Setting this to $\beta'/(2K)$ gives $e_{k,a} = \sqrt{\ln(4K/\beta')/(2n_{k,a})}$, and a union bound over the $2K$ client-groups gives the event with probability $\geq 1 - \beta'$. On it, each $|d_k - (\mu_{k,1} - \mu_{k,0})| \leq 2\max e_{k,a}$ (the clip does not bind and is 1-Lipschitz), and Lemma 1(d) transfers this to the trimmed means. □

The assumption that $\hat{\theta}$ is independent of the validation draws fails for CLAMP-RT, as discussed in the main text.

Differencing attack (closed form)

For one release, a test between $\mathcal{N}(\mu, s^2)$ and $\mathcal{N}(\mu + \Delta, s^2)$ with equal priors is optimal when it thresholds at $\mu + \Delta/2$, and its balanced accuracy is $\Phi(|\Delta|/(2s))$. With both servers honest $s = \sqrt{2}\sigma_d$; with one colluding server $s = \sigma_d$. If the target's contribution is unchanged over R releases and the attacker knows everything else, the sum of the R observations is sufficient, with mean shift $R\Delta$ and standard deviation $\sqrt{R}s$, which gives $\Phi(\sqrt{R}|\Delta|/(2s))$. With $|\Delta| \leq \Delta_{\mathrm{TM}}$ and $\sigma_d = z\Delta_{\mathrm{TM}}$ these values depend only on z.

Table S1: Falsified gap reports against CLAMP-RT: test AUC, mean±sd over 3 seeds (companion of the DPD table in the main text). $*$: $f > b$.

Dataset	K	f	Honest	hide	inflate	inlier	zero
Bail	5	1	0.781±0.046	0.781±0.047	0.781±0.046	0.781±0.045	0.781±0.046
		2*		0.764±0.038	0.775±0.038	0.781±0.046	0.781±0.046
	10	1	0.743±0.019	0.743±0.019	0.743±0.019	0.743±0.019	0.743±0.020
		2		0.743±0.019	0.743±0.019	0.743±0.019	0.743±0.019
		3*		0.732±0.020	0.743±0.017	0.744±0.019	0.744±0.019
Credit	5	1	0.764±0.005	0.766±0.004	0.765±0.005	0.766±0.004	0.766±0.004
		2*		0.734±0.004	0.629±0.042	0.767±0.004	0.767±0.005
	10	1	0.759±0.006	0.760±0.006	0.759±0.006	0.760±0.006	0.760±0.005
		2		0.760±0.005	0.759±0.005	0.760±0.005	0.760±0.005
		3*		0.760±0.008	0.668±0.035	0.760±0.005	0.761±0.005
Pokec-z	5	1	0.755±0.025	0.755±0.024	0.751±0.025	0.751±0.027	0.748±0.031
		2*		0.685±0.011	0.730±0.043	0.757±0.029	0.752±0.035
	10	1	0.768±0.002	0.768±0.003	0.768±0.003	0.768±0.003	0.768±0.004
		2		0.768±0.003	0.769±0.006	0.767±0.003	0.768±0.003
		3*		0.751±0.007	0.761±0.002	0.767±0.002	0.768±0.003
Pokec-n	5	1	0.771±0.029	0.771±0.029	0.774±0.030	0.770±0.026	0.773±0.029
		2*		0.706±0.005	0.730±0.011	0.770±0.030	0.773±0.035
	10	1	0.768±0.003	0.766±0.004	0.764±0.003	0.765±0.001	0.765±0.004
		2		0.767±0.002	0.767±0.006	0.765±0.004	0.766±0.003
		3*		0.755±0.001	0.757±0.006	0.765±0.003	0.766±0.004

Table S2: CLAMP-RT under update attacks ($K = 10$, 3 seeds): test DPD / AUC, mean±sd.

Dataset	Attack	$f = 1$	$f = 2$	$f = 3$	$f = 4$
Bail	none ($f = 0$)	DPD 0.009±0.002, AUC 0.743±0.019			
	ALIE	0.010±0.004 / 0.74	0.007±0.002 / 0.72	0.006±0.001 / 0.72	0.006±0.001 / 0.72
	Fair. poison	0.010±0.002 / 0.74	0.009±0.008 / 0.72	0.015±0.009 / 0.71	0.023±0.009 / 0.71
	IPM	0.009±0.002 / 0.73	0.006±0.002 / 0.71	0.005±0.002 / 0.71	0.004±0.001 / 0.71
	Scaling	0.009±0.002 / 0.74	0.009±0.002 / 0.74	0.008±0.002 / 0.74	0.009±0.002 / 0.74
	Sign flip	0.009±0.003 / 0.73	0.009±0.009 / 0.69	0.009±0.009 / 0.68	0.012±0.010 / 0.66
Pokec-z	none ($f = 0$)	DPD 0.046±0.008, AUC 0.768±0.003			
	ALIE	0.045±0.008 / 0.77	0.042±0.011 / 0.76	0.036±0.016 / 0.75	0.038±0.010 / 0.75
	Fair. poison	0.059±0.003 / 0.77	0.118±0.013 / 0.75	0.151±0.023 / 0.74	0.106±0.097 / 0.73
	IPM	0.045±0.009 / 0.77	0.046±0.010 / 0.78	0.044±0.008 / 0.78	0.025±0.005 / 0.76
	Scaling	0.046±0.008 / 0.77	0.046±0.008 / 0.77	0.045±0.008 / 0.77	0.046±0.008 / 0.77
	Sign flip	0.043±0.008 / 0.76	0.032±0.015 / 0.75	0.025±0.015 / 0.74	0.017±0.012 / 0.71

S2 Additional tables

Table S3: Width $h_{(K-b)} - h_{(b-f+1)}$ of the interval of Theorem 5 of the main text in the attacked release rounds with $f \leq b$ (pre-noise honest gaps): mean and maximum.

Dataset	K	mean	max
Bail	5	0.028	0.057
Bail	10	0.050	0.084
Credit	5	0.055	0.085
Credit	10	0.047	0.071
Pokec-z	5	0.044	0.072
Pokec-z	10	0.053	0.077
Pokec-n	5	0.049	0.079
Pokec-n	10	0.048	0.090

Table S4: Privacy sweep: test AUC, mean±sd over 3 seeds.

	Credit			Pokec-z			Pokec-n		
ε	$P = 1$	$P = 5$	$P = 10$	$P = 1$	$P = 5$	$P = 10$	$P = 1$	$P = 5$	$P = 10$
1	0.757±0.011	0.765±0.004	0.766±0.006	0.747±0.019	0.749±0.025	0.751±0.032	0.770±0.026	0.773±0.030	0.774±0.029
2	0.757±0.011	0.766±0.004	0.767±0.006	0.753±0.018	0.749±0.031	0.756±0.029	0.770±0.022	0.766±0.026	0.775±0.031
4	0.755±0.015	0.766±0.005	0.766±0.004	0.759±0.030	0.754±0.023	0.755±0.025	0.768±0.027	0.764±0.024	0.770±0.031
8	0.763±0.007	0.766±0.005	0.765±0.005	0.752±0.026	0.753±0.020	0.749±0.030	0.767±0.029	0.771±0.032	0.774±0.033